

Project 3: SQL Data Analysis

DecodeLabs Industrial Training Kit, Data Analytics Internship | Prepared by Sanni David Semilore

1. Dataset Overview

This analysis was performed on the same 1,200-row order dataset used in Projects 1 and 2, queried using SQL (via an in-browser SQLite-compatible engine) against a single table named `orders`, containing all 14 original columns.

2. Methodology

Five queries were written to progressively demonstrate core SQL skills: row filtering with `WHERE`, sorting with `ORDER BY`, aggregation with `COUNT`, `SUM` and `AVG`, grouping with `GROUP BY`, and filtering aggregated results with `HAVING`. Each query and its result are documented below.

3. Query 1: Filtering with WHERE

```
SELECT * FROM orders WHERE Product = 'Laptop';
```

This query isolates all orders for a single product. Result: 173 rows returned out of 1,200 total orders, confirming Laptop orders make up roughly 14.4% of the dataset.

4. Query 2: Numeric Filtering with ORDER BY

```
SELECT OrderID, Product, TotalPrice FROM orders WHERE TotalPrice > 2000 ORDER BY TotalPrice DESC;
```

This finds high-value orders and sorts them from highest to lowest. Result: 180 rows returned, with the highest single order being ORD200789 (Tablet, 3,456.40), matching the maximum value identified independently in the Project 2 EDA.

5. Query 3: Aggregation with GROUP BY

```
SELECT Product, COUNT(*) AS OrderCount, ROUND(SUM(TotalPrice),2) AS TotalRevenue, ROUND(AVG(TotalPrice),2) AS AvgOrderValue FROM orders GROUP BY Product ORDER BY TotalRevenue DESC;
```

Product	Order Count	Total Revenue	Avg Order Value
Chair	178	195,620.11	1,098.99
Printer	181	195,612.61	1,080.73
Laptop	173	192,126.56	1,110.56
Tablet	179	186,568.95	1,042.28
Monitor	163	175,651.41	1,077.62

Desk	170	167,459.93	985.06
Phone	156	151,722.39	972.58

Chair generates the highest total revenue, driven largely by order volume (178 orders) rather than order size. Laptop, by contrast, has the fewest orders among the top four products but the highest average order value, meaning each Laptop order tends to be worth more individually.

6. Query 4: Filtering Grouped Results with HAVING

```
SELECT Product, ROUND(AVG(TotalPrice),2) AS AvgOrderValue FROM orders GROUP BY Product
HAVING AVG(TotalPrice) > 1050 ORDER BY AvgOrderValue DESC;
```

Product	Avg Order Value
Laptop	1,110.56
Chair	1,098.99
Printer	1,080.73
Monitor	1,077.62

Unlike WHERE, which filters individual rows before grouping, HAVING filters the grouped results after aggregation. Here, only 4 of the 7 products (Laptop, Chair, Printer, Monitor) have an average order value above 1,050; Tablet, Desk and Phone were excluded.

7. Query 5: Top Customers by Spend

```
SELECT CustomerID, COUNT(*) AS NumOrders, ROUND(SUM(TotalPrice),2) AS TotalSpent FROM
orders GROUP BY CustomerID ORDER BY TotalSpent DESC LIMIT 5;
```

CustomerID	Num Orders	Total Spent
C38840	2	5,723.23
C57276	1	3,456.40
C67260	1	3,390.80
C13877	1	3,384.90
C18404	1	3,370.20

The top customer, C38840, reached the highest total spend through 2 separate orders, while the remaining top 5 customers each reached their spend through a single high-value order. This distinguishes genuine repeat-customer value from one-off large purchases.

8. Key Observations

- Sales are fairly evenly distributed across products, with no single product accounting for a dominant share of revenue.
- Laptop has the highest average order value despite having fewer orders than most other products, indicating customers spend more per Laptop purchase on average.
- Chair leads in total revenue primarily through order volume rather than order size.
- HAVING is essential for filtering on aggregated metrics (like average order value) since WHERE cannot filter on values computed after grouping.
- The top customer by total spend achieved that position through repeat purchases, while other high-spending customers reached similar totals through a single large order.

9. Conclusion

This analysis demonstrates the core SQL skills required for Project 3: filtering rows with WHERE, sorting with ORDER BY, aggregating with COUNT, SUM and AVG, grouping with GROUP BY, and filtering aggregated results with HAVING. The findings are consistent with the Excel-based EDA performed in Project 2, confirming that both approaches, spreadsheet formulas and structured queries, arrive at the same underlying business insights.